

PROJECT MANAGEMENT SYSTEM (PMS)

1. SYSTEM OVERVIEW

The Project Management System (PMS) is a role-based management platform designed to streamline project and task handling within organizations. The system supports multiple roles with distinct privileges:

- **ADMIN:** Full system control, user and department management, analytics, and system configuration.
- **TEAM LEAD:** Project creation, task assignment, observer management, and tracking team performance.
- **EMPLOYEE:** Task execution, time tracking, and personal project management.

PMS enables efficient collaboration, automated notifications, and accurate time tracking for projects and tasks.

2. USER ROLES & ACCESS

2.1 ADMIN

- Create/Edit/Delete users: ✓
- Manage Departments & Designations: ✓
- View all projects and tasks: ✓
- System-wide analytics: ✓
- User activation/deactivation: ✓

2.2 TEAM LEAD

- Create projects: ✓
- Assign tasks to employees: ✓
- Add observers to tasks: ✓
- View team members: ✓
- Track project progress: ✓

2.3 EMPLOYEE

- View assigned tasks: ✓

- Start/Pause/Resume/Complete tasks: ✓
 - Add task summaries: ✓
 - View personal projects: ✓
 - Track personal time: ✓
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3. MODULES & FEATURES IMPLEMENTED

3.1 Authentication Module

- Login with Email: ✓
- Role-based redirect: ✓
- Inactive user block: ✓
- Session management: ✓
- Logout: ✓

3.2 Dashboard Module

- Welcome header with stats: ✓
- Statistics cards: ✓
- Quick actions: ✓
- Recent projects (Admin/TL): ✓
- Recent tasks: ✓
- Team members grid (TL only): ✓
- User management table (Admin only): ✓

3.3 Project Management

- Create/Edit/Delete projects: ✓
- View all projects and details: ✓
- Multiple assignees: ✓
- Add project resources: ✓

3.4 Task Management

- Create/Edit/Delete tasks: ✓
- Task details and list view: ✓
- Multiple assignees & observers: ✓
- Estimated time & deadline: ✓

3.5 Time Tracking

- Start/Pause/Resume/Complete tasks: ✓

- Live timer & total time display: ✓
- Estimated time display: ✓
- Timezone support: ✓

3.6 Task Summary

- Summary required before completion: ✓
- Add/view summaries: ✓
- Completion verification: ✓

3.7 Notification System

- Task assignment/start/completion notifications: ✓
- Notification bell with unread count: ✓
- Dropdown preview & full notification page: ✓
- Mark as read: ✓

3.8 User Management (Admin)

- Create/Edit/Delete users: ✓
- Activate/Deactivate users: ✓
- Department and designation management: ✓

3.9 Team Lead Features

- View team members and tasks: ✓
- Assign tasks and create projects: ✓
- Team statistics: ✓

3.10 Employee Features

- My Tasks & My Projects view: ✓
- Time tracking: ✓
- Task summary management: ✓
- Personal dashboard: ✓

4. TESTING SCENARIOS

4.1 Login Testing

- Valid login: Redirect to role dashboard
- Invalid password/non-existent email: Error message

- Inactive user: "Account inactive"
- Empty fields: Field required message

4.2 Admin Testing

- User creation/edit/deletion: Reflected in list
- Deactivation: User cannot login
- Department creation/view: Listed and saved

4.3 Project Testing

- Project create/edit/delete: Saved/removed with confirmation
- Add resources: Appears in project details

4.4 Task Testing

- Task create/assign/add observers: Notifications triggered
- Set estimated time: Displayed correctly
- Task list view: Table shows all tasks

4.5 Time Tracking Testing

- Start/Pause/Resume/Complete: Timer accurate
- Multiple pauses: Total time correct

4.6 Task Summary Testing

- Completion without summary: Redirects to summary
- Add summary: Saved successfully
- Complete with summary: Task completes
- View summary: Displayed in task details

4.7 Notification Testing

- Task assignment/start/completion notifications: Sent to respective roles
- Read notification: Marks as read
- Unread count: Correctly displayed

This report summarizes the Project Management System (PMS), its modules, user roles, implemented features, and the comprehensive testing scenarios executed to ensure system stability and functionality.